

ODD Protocol: Religious and Linguistic Segregation Model in Karachi

Modeling Social Complexities — Fall 2025
MSc Final Project
Hussain Mustansir — M Hassaan Tariq

December 15, 2025

Overview

The model description follows the ODD (Overview, Design concepts, Details) protocol for describing individual- and agent-based models (Grimm et al. 2006; 2010; Railsback and Grimm 2018).

1 Purpose and Patterns

Purpose

This model examines how religious and linguistic differences shape residential patterns in Karachi, Pakistan. Specifically, it investigates whether a **global** tolerance threshold, when applied to household migration decisions, can drive region-level segregation patterns similar to those observed in census data.

The model addresses three primary questions:

1. How do global tolerance thresholds for religious and linguistic similarity affect household migration flows?
2. Can decentralized avoidance behaviors produce observable segregation patterns at the administrative region level?
3. How do empirically derived birth and death rates influence demographic composition over a projected 100-year period (2025–2125)?

Patterns

The model utilizes empirical census data from 2017 as the baseline for religious and linguistic distributions across Karachi’s administrative regions (ADM3 level). The model is considered useful if it can:

- Maintain demographic distributions approximating the initial census data when tolerance thresholds are high (low segregation pressure).
- Generate regional concentration patterns where specific religious or linguistic groups cluster in specific administrative units when tolerance thresholds are reduced.

2 Entities, State Variables, and Scales

Entities

The model includes four distinct types of entities:

1. Households (Agents) Households are the primary mobile agents. They possess specific cultural identities and make migration decisions. State variables include:

- **religion:** Categorical identity (Muslim, Hindu, Christian, Other).
- **language:** Categorical identity (Urdu, Sindhi, Pashto, Other).
- **my-region:** The name of the ADM3 administrative unit the agent currently resides in.
- **my-manager:** A reference to the Manager agent responsible for the current region.

2. Managers To optimize computational efficiency, the model utilizes "Manager" agents. Instead of households scanning thousands of neighbors (an $O(N^2)$ or spatial indexing operation), each region has one Manager that aggregates demographic data. This allows households to query regional statistics in $O(1)$ time. Manager state variables include:

- **m-region-name:** Name of the region managed.
- **m-total:** Total household count in the region.
- **Demographic counters:** **m-muslim**, **m-hindu**, **m-urdu**, **m-sindhi**, etc.

3. Patches Patches represent geographic locations derived from the GIS shapefile. State variables include:

- **p-region-name:** The administrative region the patch belongs to.
- **is-habitable?:** Boolean flag indicating if the patch is within valid administrative boundaries.

4. Global Environment Global variables include the GIS dataset (**pak-dataset**), the agent scaling factor (**agent-scale**), and user-defined parameters for tolerance and visualization.

Scales

- **Spatial Scale:** The model covers Karachi at the ADM3 (Tehsil) administrative level using a GIS shapefile.
- **Temporal Scale:** One time step (tick) represents one year. The simulation runs for 100 time steps, representing the period from 2025 to 2125.
- **Agent Scale:** 1 Agent represents 5,000 people (defined by **agent-scale** = 5000).

3 Process Overview and Scheduling

The model follows this schedule within each time step (Year):

1. **Update Manager Statistics:** Managers reset counters. Every household reports its identity to its assigned manager. Managers aggregate these counts to produce a regional "Census."
2. **Check Happiness and Move:** Households query their manager to calculate the proportion of their own group in the region. If this proportion is below the global tolerance threshold, the household relocates immediately.
3. **Vital Dynamics:** Births and deaths occur probabilistically based on national rates.
4. **Update Visualization:** Agents and regions are recolored based on the selected visualization mode (**viz-mode**).

4 Design Concepts

Basic Principles

The model extends Schelling’s segregation model to a regional level using dual identity dimensions (Religion and Language). The core principle is ”threshold-based sorting”: households are satisfied only if their cultural group constitutes a minimum percentage of their region’s population.

Sensing and Interaction (The ”Manager” System)

Direct agent-to-agent sensing and interaction is replaced by agent-to-manager interaction.

- **Justification:** Calculating neighborhood composition for thousands of agents normally requires expensive spatial searches. By introducing Managers, the model reduces the complexity of the ”sensing” phase from quadratic to linear time ($O(N)$), allowing for larger population scales.

Adaptation

Households utilize a simple adaptive behavioral rule to maximize satisfaction. If a household’s cultural group (religion or language) falls below the global tolerance threshold in their current region, they adapt by relocating to a randomly selected alternative region.

Stochasticity

Stochasticity is used in:

- **Initialization:** Agents are assigned attributes probabilistically based on census data.
- **Relocation:** Dissatisfied agents choose a *random* destination region, rather than optimizing for the ”best” region. This reflects imperfect information and limited housing search capacity.
- **Vital Dynamics:** Births and deaths are probabilistic events.

Observation

The visualization allows the user to toggle between ”Religion” and ”Language” modes.

- **Focus Groups:** Users can highlight specific groups (e.g., Christians). Minority groups are visualized using a shifted color scale to ensure their regional impact is visible.

5 Initialization

The model initializes as follows (`setup2`):

1. **Load GIS Data:** The shapefile `karachi_census_merged_fixed.shp` is loaded. Patches are mapped to regions.
2. **Create Managers:** One manager is created for each unique region name found in the dataset.
3. **Spawn Agents:**
 - The number of agents for a region is calculated as $\lfloor Total_Pop/5000 \rfloor$.
 - Agents are placed on ”interior” patches (patches with 8 same-region neighbors) first to prevent edge artifacts.
 - Attributes (Religion/Language) are assigned probabilistically using cumulative probability derived from the region’s census attributes (e.g., `Muslim_Pop`, `Urdu_Pop`).

6 Input Data

The model relies on a processed GIS shapefile containing ADM3 boundaries and 2017 Census demographic data.

- **Source:** Humanitarian Data Exchange (humdata.org), processed via Python (Geopandas) to merge census CSV data.
- **Attributes:** Due to shapefile character limits, some attributes are truncated (e.g., `Christian_Pop` is stored as `ChristianP`). The model code handles these truncated field names correctly.

7 Outputs

The model generates real-time visual and numerical outputs to facilitate Exploratory Data Analysis (EDA).

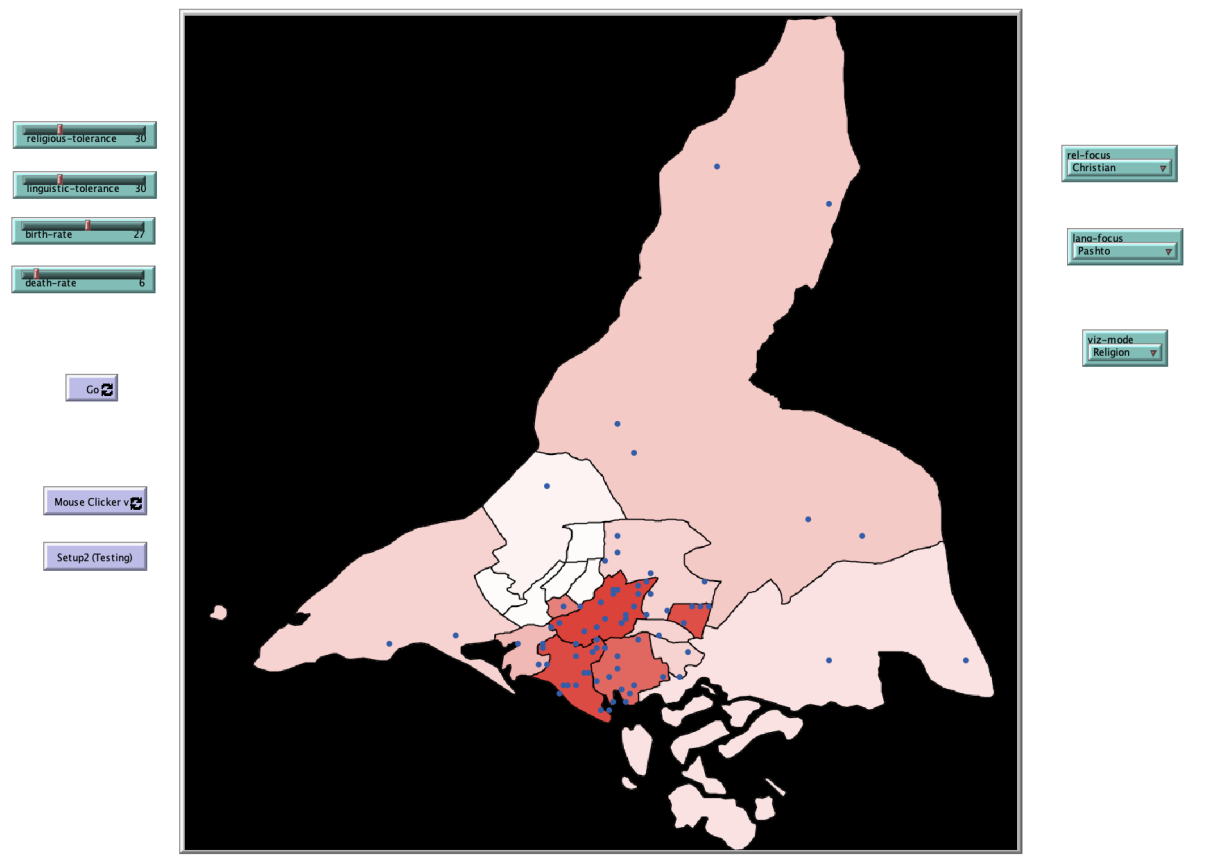


Figure 1: **Geospatial Visualization:** The GIS map of Karachi (ADM3) colored by dominant religious group. White areas represent uninhabited zones or regions with insufficient data.

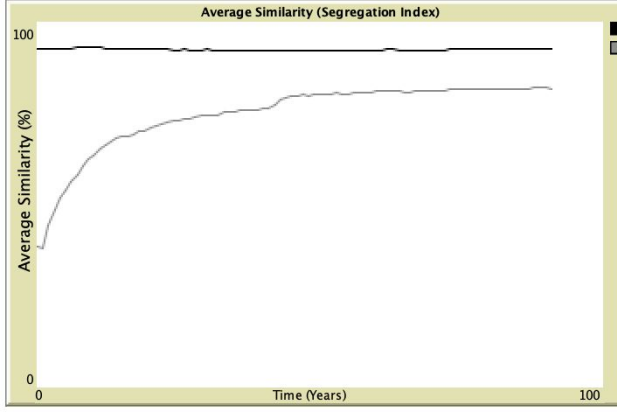


Figure 2: **Average Similarity Index:** Tracks the global mean of household satisfaction (percent of same-group neighbors).

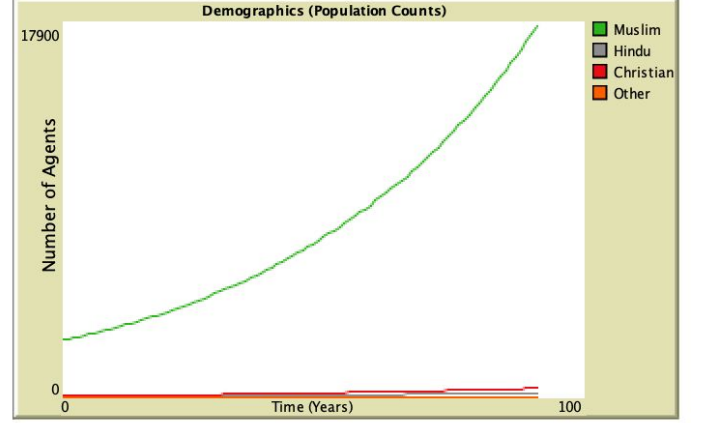


Figure 3: **Population Demographics:** Real-time tracker of total population counts for each religious group.

- **Geospatial Visualization:** As shown in Figure 1, the map updates annually to reveal clustering patterns.
- **Time-Series Plots:** The plots above track the emergence of segregation and demographic shifts due to vital dynamics.

8 Submodels

Update Manager Stats

Runs at the start of every tick.

- Managers reset all counters to 0.
- Each household asks `my-manager` to increment the counter corresponding to its `religion` and `language`.
- This ensures decision-making is based on the *current* year's population distribution.

Check Happiness and Move

Each household calculates two ratios based on data from `my-manager`:

$$R_{rel} = \frac{\text{Count of my religion in region}}{\text{Total households in region}} \times 100$$

$$R_{lang} = \frac{\text{Count of my language in region}}{\text{Total households in region}} \times 100$$

The household is "unhappy" if:

$$R_{rel} < \text{religious-tolerance} \quad \text{OR} \quad R_{lang} < \text{linguistic-tolerance}$$

Note: Tolerance sliders range from 0 to 100. Default setting is 30 (30%).

If unhappy, the household calls `relocate`.

Relocate

1. The agent identifies all regions *except* its current one.
2. It selects one target region randomly.
3. It moves to a random habitable patch within that target region.
4. It updates `my-region` and links to the new `my-manager`.

Vital Dynamics

Applied to every household annually:

- **Death:** With probability $P = \text{death-rate}/1000$, the agent dies (is removed). Default slider value: 6.0 (0.6%).
- **Birth:** With probability $P = \text{birth-rate}/1000$, the agent hatches a new agent. The new agent inherits the parent's location, religion, and language. Default slider value: 27.0 (2.7%).

Update Visualization

Updates the agent color based on `viz-mode`:

- **Religion Mode:** Muslim (Green), Hindu (Orange), Christian (Blue), Other (Grey).
- **Language Mode:** Urdu (Blue), Sindhi (Yellow), Pashto (Red), Other (Grey).

These are tentative colors and can be easily changed as per preferences in the `get-color-for-religion` and `get-color-for-religion` procedures.

References

1. Grimm, V. et al. (2006). A standard protocol for describing individual-based and agent-based models. *Ecological Modelling*, 198(1-2), 115-126.
2. Railsback, S. F., & Grimm, V. (2019). *Agent-Based and Individual-Based Modeling: A Practical Introduction*. Princeton University Press.
3. Pakistan Bureau of Statistics. (2017). *Population Census 2017.s*
4. <https://data.humdata.org/dataset/geoboundaries-admin-boundaries-for-pakistan> (Shapefile related data)